Language Features under Weight Quantization: Activation Agreement and Intervention Effects

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Abstract

How much does preservation of a sparse autoencoder (SAE) activation tell us about preservation of its effect on a language model? We investigate 27 languages using Gemma-2-2B, a fixed Gemma Scope dictionary, two round-to-nearest (RTN) implementations and GPTQ, with extensions to Gemma-2-9B and Llama-3.2-1B. Removing selected language-feature groups raises own-language next-token loss beyond activity-matched controls in all 27 languages of both Gemma sizes. Their effects remain near full precision at six bits and increase at four bits under RTN; GPTQ attenuates this increase. Activation agreement, language-association retention and intervention-effect similarity are distinct: at legacy RTN INT4, only 13.2% of a 1,487-feature sample passes the activation-survival cutoff, yet median effect similarity is 0.839 and feature-identity retrieval is 99.7%. Continuous activation correlation adds a small positive contribution to language-held-out effect prediction, reducing error by 3.1–4.4%. An exact gain/noise decomposition clarifies its scale dependence. Crossed context–weight experiments show quantizer-dependent intervention transfer, with INT4 effect cosine improving from 0.768 under legacy RTN to 0.877 under GPTQ. A held-out comprehension intervention has no established full-precision effect, and patching the selected groups recovers little total quantization loss. Together these results support useful, bounded causal evidence from SAE interventions and a practical distinction between activation preservation and intervention portability. Most analyses are exploratory; the original held-out Belebele family is confirmatory.

1 Introduction

Weight quantization can change multilingual language-model behavior unevenly (Marchisio et al., 2024). Sparse autoencoders (SAEs) offer an additional view by expressing internal activations as sparse combinations of learned directions (Cunningham et al., 2023). Language-associated features can be selected and manipulated (Deng et al., 2025), and activation correlation before and after quantization measures one form of preservation (Duan, 2026). The unresolved empirical question is how these observations relate to the effects of intervening on the corresponding features.

Several outcomes can differ without contradiction. A feature can change its activation magnitude while retaining a preferred language. An intervention can retain a recognizable output effect while its precise direction moves. A group can matter for predicting text without producing a measurable change on a particular comprehension endpoint. Evaluating these distinctions gives a more useful account of compression than treating every changed coordinate as a lost mechanism.

We report the completed investigation through four *retrospective synthesis questions*. They organize the accumulated evidence and do not replace the original frozen hypotheses: **RQ1**, how do language selectivity and resource tier relate to activation preservation across quantizers? **RQ2**, what do language-group interventions change across tasks and precision? **RQ3**, how informative are activation survival and association retention about intervention-effect preservation? **RQ4**, how do context, downstream weights and generic sensitivity affect transfer of a fixed intervention?

Our contribution is a controlled empirical evaluation with activity-matched groups, donor patches, individual effect fingerprints, a crossed prefix/suffix design, two corpus panels, and a streamed measurement decomposition. The Gemma-2-9B extension tests the group pattern at another size; Llama supplies an informative contrasting model/dictionary setting. These experiments establish effects of defined interventions in the tested systems. Generalization and a complete account of natural language mechanisms are separate questions.

2 Related Work and Measurement Context

Multilingual compression. Marchisio et al. (2024) show that quantization degradation depends on language, task and evaluation. Resource tiers describe a broader inclusion problem (Joshi et al., 2020), but tiers do not experimentally manipulate training exposure. GPTQ uses approximate second-order information and calibration data to improve quantization (Frantar et al., 2022). Our matched-RTN arm distinguishes changes in grid and scale granularity from the remaining combined contribution of calibration and compensation.

SAE features and their evaluation. Gemma Scope supplies open SAEs (Lieberum et al., 2024); JumpReLU and TopK improve reconstruction/sparsity tradeoffs (Rajamanoharan et al., 2024; Gao et al., 2024). These improvements coexist with shared questions about feature completeness and interpretation. Leask et al. (2025) show that SAE units need not be canonical, while Makelov et al. (2025) evaluate task-grounded control without requiring exact feature recovery. SAEBench finds that proxy gains need not produce practical improvements (Karvonen et al., 2025). These studies motivate explicit functional endpoints and controls. Our fixed dictionary supplies an operational basis for paired comparisons; useful intervention evidence does not require claiming that its coordinates uniquely recover every natural computation.

Preservation and intervention. Deng et al. (2025) already show language-feature ablation and steering, so FP group effects are supporting evidence rather than a first-ever claim. Duan (2026) study activation survival under RTN; Borobia et al. (2026) distinguish geometric survival and importance under pruning; Hoang et al. (2026) examine the geometry of feature effects. Chang et al. (2025) use activation patching to localize quantization error, and Vaidyanathan et al. (2026) analyze interactions in patching. We connect activation, association and effect measurements across quantizers and test intervention transfer with explicit control directions. Belebele (Bandarkar et al., 2024) supplies a separate reading-comprehension endpoint.

3 Experimental Setting and Status

Models and dictionaries. The primary model is Gemma-2-2B (Gemma Team et al., 2024), with the layer-12 post-block residual Gemma Scope JumpReLU SAE (16,384 features). Inference uses bfloat16, SAE computations and edits float32. The completed original-RTN Gemma-2-9B extension uses its own 16k Gemma Scope SAE at layer 20 and the same language/sentence recipe. The contrasting Llama-3.2-1B setting (Grattafiori et al., 2024) uses a 14,336-feature TopK SAE ($k = 48$), layer 12 of 16, with different training data and tokenizer. Comparisons between these setups do not isolate any one architectural difference.

Quantizers and precision. Legacy RTN is the project's reference implementation; attention scales share an output-coordinate scale across heads. Matched RTN uses a per-output-channel layout and the same grid as GPTQ. GPTQ calibrates on 8,192 Wikipedia tokens per language (221,184 total), disjoint from FLORES sentences and external evaluation articles. Selected transformer weights are rounded, with embeddings and normalization parameters unchanged; values remain in floating-point storage. We evaluate weight transformations, not integer-kernel speed or memory. The original ladder is FP, INT8–INT4, plus INT16/INT12 perturbation references and an INT0 restore identity. New quantizers use INT4/INT6 throughout, with INT8–INT4 in the activation stage. INT16 makes a small real weight change; it is a reference scale, not pure numerical noise.

Language and task panels. The 27 parallel FLORES-200 (NLLB Team et al., 2022) languages comprise nine per descriptive resource tier (Appendix A). Selection uses dev. Association evaluation uses 27,036 content tokens per language; functional loss uses 300 aligned devtest sentences per language. One BOS token is prepended and excluded, with padding, from statistics and edits. Belebele calibration selects a letter-answer format and 18 languages above chance plus 0.05. Evaluation has 385 questions per language in 219 passage clusters. A separate Wikipedia transfer panel has six languages, fresh articles, and fixed feature selections.

Analysis status and reproducibility. FLORES functional analyses reuse previously examined sentences and remain exploratory, including stages with choices frozen before their own outputs. The only fully confirmatory family is the original held-out Belebele family. No verified external preregistration is claimed. The final RQs are synthesis, not newly confirmed hypotheses. An audited ini-

tial pipeline was retired. Frozen code, manifests, sham/restore identities, input hashes and analysis reproduction checks support implementation integrity. Both new quantizers reproduce the FP values and reference conditions exactly; the 9B analysis reproduces its imported tables byte-identically from pinned raw files. This is analysis reproduction rather than independent repeated inference.

4 Measurements and Controls

Activation and association. For feature f, activation survival uses $R_1 = \mathrm{corr}_t(a_f^{\mathrm{FP}}(t), a_f^b(t))$ and the historical threshold $R_1 > 0.9$. Language selectivity uses $S_f = \max_\ell r_{\ell f} / \operatorname{mean}_\ell r_{\ell f}$ from firing rates, a shuffled-language cutoff (50 permutations), support thresholds and stable peaks in 80 of 100 resamples. The 2B cohort has 8,086 selected features. A feature establishes its language on devtest when the original language is its unique supported peak; association retention conditions on that FP establishment. Thus activation correlation and association retention have different populations and meanings.

Interventions and net harm. G_ℓ contains the top 64 selective features by dev activation mass. Controls match log firing rate and log mean activation from 2,860 supported non-selective features, choosing among nearby unused candidates (three K64 control draws in the evaluations, from five available in the manifest). Matching is imperfect: 75/81 language–size ratios meet $[0.8, 1.25]$, with Thai, Yoruba and Lao outside at K64. The comparison estimates excess harm against these controls, not exclusive language specificity. For residual x, activation a_f and decoder d_f, edits preserve the reconstruction error:

$$x' = x + \sum_{f \in G} (a'_f - a_f) d_f. \tag{1}$$

Zero ablation sets $a'_f = 0$, graded scaling sets $a'_f = \alpha a_f$, and donor patches substitute the FP activation at the same position. Harm is edited minus unedited NLL within the same model state. Excess harm E_ℓ^b subtracts mean control harm. Retention is $E_\ell^b / E_\ell^{\mathrm{FP}}$ where FP effect is established; a mean of ratios, median ratio and ratio of mean effects are distinct estimands. Donor recovery divides the patch's loss improvement by the quantization-induced loss gap. These patches are nonadditive and do not partition causal responsibility.

Effect similarity and uncertainty. W_1 is cosine similarity of a single-feature ablation's output-effect fingerprint with its FP fingerprint, with different-feature and direction controls. The F5 sample has 1,487 stratified features; 1,412 meet the INT16 reference criterion for reliable effects. The INT16 cosine's fifth percentile (0.9489) supplies a comparative fidelity convention. This is not a semantic correctness threshold. Loss intervals use 2,000 paired sentence-cluster resamples, Belebele passage resamples, and F5 language resamples. Transfer uses crossed feature/source-cluster resamples. The reviewed 9B comparisons additionally report 5,000 language-bootstrap intervals. Intervals condition on their samples and fixed feature/control choices; exploratory intervals are not multiplicity-adjusted confirmation.

5 RQ1: Activation Preservation across Quantizers

Initial contrasts and resource tiers. The published-regime WikiText reproduction closely matches the reference table (INT7 81.19% versus 81.26%, INT6 51.15% versus 51.30%). Adding BOS changes FP perplexity from 458.4 to about 16.1 and INT6 survival to about 97.5%, demonstrating regime sensitivity. In the adopted BOS-masked regime, none of three original tests rejects after Holm correction: selective-minus-control survival is -0.28 and -1.44 percentage points at INT7/6 (adjusted $p = 0.514, 0.128$). Non-rejection is not equivalence. A stricter post hoc selectivity cutoff ($S > 5$) gives -7.16 points at INT6, which reverses to $+4.58$ after matching in ten activation-magnitude bins. Tier survival at INT6 is 69.4/90.6/95.1% (low/mid/high); its fitted tier slope attenuates from $+12.3$ to -0.81 points after scale adjustment. This is an observational adjustment, not an identified training-resource mechanism.

The quantizer changes preservation. At INT6, average all-active survival is 85.0/89.7/97.5% for legacy/matched/GPTQ; INT4 values appear in Table 1. GPTQ exceeds legacy RTN at INT4 by 17.83 survival points $[16.29, 19.38]$, and matched RTN by 11.86 $[11.18, 12.50]$, with paired language-bootstrap intervals. Relative to legacy RTN it lowers mean NLL increase by 0.461 nats/token $[0.348, 0.595]$, improving both metrics in all 27 languages. Matched RTN alone closes part of the gap. Its small selective contrasts cross the original within-family significance threshold, while

INT4 measurement	Legacy	Matched	GPTQ
Active-feature survival (%)	12.7	18.6	30.5
Mean loss increase (nats/token)	0.590	0.385	0.130
Median group-effect ratio	1.54	1.36	1.22
Median effect cosine (W_1)	0.839	0.875	0.923
Transfer cosine	0.768	0.822	0.877
Belebele accuracy (%)	23.6	29.1	31.8

Table 1: Completed Gemma-2-2B quantizer comparison. Rows have distinct populations and estimands: survival averages per-language all-active shares, W_1 uses F5 features, transfer uses eligible feature–context items, and Belebele uses 18 languages.

GPTQ's do not; those reruns are exploratory additional families, not new confirmation. The result is quantizer-dependent multilingual preservation, with no robust quantizer-independent selective penalty established.

An exact measurement decomposition. For paired activations a, b, write the least-squares decomposition $b = ga + c + e$, with residual orthogonal to a. Then, on nondegenerate cells,

$$\lambda = \log(R_1^{-2} - 1) = \log \sigma_e^2 - 2 \log |g| - 2 \log \sigma_a. \tag{2}$$

The survival threshold is $\lambda < -1.450$. Newly streamed sufficient statistics reproduce prior correlations within 3×10^{-8} and the identity to about 10^{-14} across 214,506 FP-active language–feature cells. Threshold-normalized encoder perturbation ν measures RMS encoder preactivation change on FP-active tokens divided by the feature threshold. Across six INT4/6 cells, median Spearman$(\lambda, \log \sigma_a)$ is -0.507 to -0.534, whereas the corresponding ν correlations are -0.004 to 0.114. Thus survival partly reflects signal scale; ν offers a different, more scale-separated diagnostic. RTN tier gradients also occur at INT16; GPTQ tier estimates change with scale adjustment. These results refine measurement interpretation without establishing either selective circuit destruction or an absence of language disparities.

6 RQ2: Causal Relevance across Tasks and Precision

Group ablation and graded dependence. At FP the 2B K64 excess harm is positive in all 27 languages, with mean 0.704 nats/token $[0.684, 0.723]$ and median 0.262. Retention stays near one through INT6, then increases: legacy INT4 median ratio 1.54 $[1.43, 1.62]$, macro mean 2.14 $[2.04, 2.50]$, ratio of means 1.33. GPTQ reduces the median ratio to 1.22 $[1.14, 1.24]$; its macro mean 1.31 $[1.27, 1.50]$ receives an inconclusive classification against the frozen $[0.75, 1.33]$ band. The point estimates show increased sensitivity, not improved language ability or a location of quantization damage.

Study G gives converging intervention evidence. Harm rises with deeper suppression in all 27 languages; median FP excess harm is 0.0223/0.0902/0.1687/0.2624 at multipliers 0.354/0.177/0.088/0. The strict precision-monotonicity rule fails under legacy RTN because of a small intermediate-width dip, while the dose-response and consistency rules hold. Main fine-gain repair criteria fail. A fresh-sentence legacy follow-up finds a small net improvement at $1.1\times$ (-0.00135 nats/token, 97.5% interval $[-0.00197, -0.00032]$, 20/27 negative), but not at $1.2\times$ by its rule. The fresh-sentence follow-up was legacy-only. In the main 300-sentence comparison, GPTQ's best-gain-above-one count is 7/27, the same as FP. State-defined groups overlap FP groups less at INT4 (Jaccard 0.488 versus 0.855 at INT6) but are not more consequential (median difference -0.0098). Optimal gain is reliable within the study yet not associated with inverse shrinkage ($\rho = -0.17$, $p = 0.804$).

Evidence at another Gemma size and a contrasting setup. Reviewed 9B original-RTN results reproduce from pinned raw files. FP excess harm is positive in all 27 languages, median 0.270 $[0.197, 0.608]$. Language-bootstrap median retention is 1.019 $[0.978, 1.045]$ at INT6 and 1.586 $[1.238, 2.264]$ at INT4. INT4 macro mean 2.479 and ratio of mean effects 1.474 show why the aggregation must be named. Paired 2B–9B difference intervals include zero, supporting a similar ladder without a size effect. The original prediction record contains a reversed dose-order inequality and a selectivity criterion guaranteed by selection; those two items are not independent confirmations. The direct dose-response is ordered in all 27 languages. No 9B GPTQ result is included.

Llama gives a positive but smaller FP median effect, 0.048 $[0.018, 0.173]$, with 21/27 positive rather than the required 24. Its frozen consistency verdict is therefore not met. Its INT4 median ratio is 0.86 $[0.74, 1.03]$, and dose ordering is less

consistent. This establishes heterogeneity across the two model/dictionary setups while leaving the Gemma intervention contrasts intact.

Comprehension and donor recovery. On held-out Belebele the FP excess correct-option loss is -0.0001 nats/item $[-0.0035, +0.0031]$. This endpoint is distinct from accuracy: 0.426 unedited, 0.426 group-ablated, 0.425 control-ablated. The required FP effect is not established, so the five conditional retention statements are undefined. Whole-group ablation agrees; non-selective control sets of 1,024–2,860 features lower accuracy by 1.2–2.2 points, with other-language selective controls reaching 6.0 points, and sham edits change nothing. Under GPTQ, unedited INT4 accuracy is 0.318, above chance (0.25), while the small group-accuracy change persists. The null is task-specific and does not negate the next-token effects.

At INT4, FP donor patches recover 2.6% $[2.4, 2.8]$ of the legacy RTN loss gap via K64, compared with 39.3% $[38.5, 40.0]$ via the full residual. GPTQ gives 2.0% $[1.6, 2.3]$ and 41.4% $[40.2, 42.4]$. These measured recovery shares show limited efficacy of this repair, rather than a bound on the features' causal contribution.

7 RQ3: Activation Agreement and Functional Preservation

Only 6,663 of 8,086 selected 2B features establish their peak language on FP devtest. Of those, 70.8% retain it at legacy INT4, whereas correlation survival in that established cohort is 13.75%. A small FP peak margin predicts association loss (within-language AUROC 0.90/0.81/0.75 at INT6/5/4), consistent with near ties making labels sensitive. Matched lost/retained subgroup interventions do not show a systematic harm gap; median differences are $+0.0006, +0.0007, +0.0002$ nats/token at INT4/5/6. These findings distinguish association status from intervention effects.

The F5 sample provides a positive functional result alongside measured drift. At legacy INT4 only 13.2% pass activation correlation 0.9, yet median effect cosine is 0.839 $[0.829, 0.848]$, 99.9% exceed different-feature controls and 99.7% retrieve their own FP identity. Just 0.3% meet the stricter INT16-derived fidelity convention. Thus identifiable function and exact preservation coexist at different levels. GPTQ improves median INT4 effect cosine to 0.923 and the share meeting that convention to 11.7%. Binary association retention does not reliably discriminate this preservation (legacy descriptive AUROC 0.506; GPTQ 0.49 with sufficient events).

Continuous R_1 is more informative. On 1,412 reliable features, leave-one-language-out ridge prediction of W_1 controls for activity, contexts and FP effect size. Adding R_1 reduces MAE by 3.1–4.4% across all six quantizer/width cells, with every 95% language-bootstrap interval above zero. Legacy INT4 improvement is 3.7% $[2.3, 5.2]$, GPTQ 3.5% $[2.1, 5.0]$. Partial rank correlations are 0.239–0.399. These are modest positive contributions, below the project's 10% practical target; the target does not turn them into zero information. Adding gain and $\log \nu$ changes prediction by -0.4% to $+1.8\%$, with all intervals including zero. In Llama's 610 reliable features the INT4 association repeats (partial $\rho = 0.252$ $[0.184, 0.322]$, MAE gain 4.2% $[1.6, 6.7]$); INT6 is unresolved. The gain/noise diagnostics clarify activation changes without demonstrating further predictive benefit here.

8 RQ4: Portability of Fixed Interventions

Crossed design. A fixed edit subtracts the FP decoder contribution at FP-active positions. Let $D[p, s]$ be its centered output-logit effect for prefix p and suffix s, each FP or quantized Q, subtracting the cell's own unedited baseline. Define

$$C = D[\mathrm{Q}, \mathrm{FP}] - D[\mathrm{FP}, \mathrm{FP}], \tag{3}$$

$$W = D[\mathrm{FP}, \mathrm{Q}] - D[\mathrm{FP}, \mathrm{FP}], \tag{4}$$

$$I = D[\mathrm{Q}, \mathrm{Q}] - D[\mathrm{Q}, \mathrm{FP}] - D[\mathrm{FP}, \mathrm{Q}] + D[\mathrm{FP}, \mathrm{FP}]. \tag{5}$$

The total vector change equals $C + W + I$ exactly. Norms, medians and projected loss changes need not add. A fifth cell uses the quantized state's own amplitude. The panel contains 404 features and 6,464 pairs of features and contexts, with 2,203 eligible real-direction items (235 features) whose FP effect exceeds five INT16 reference floors. Matched swapped decoder rows and three random norm-matched directions control generic sensitivity. Feature hashes split calibration and held-out features; contexts can overlap between those halves.

Context and downstream weights. Legacy INT4 fixed group net harm is 0.848 versus 0.263 in FP; median absolute context/weight terms are 0.310/0.104 nats/token. Native-amplitude harm

is 0.433. The paired native-minus-fixed difference is negative in every language (median -0.351 $[-0.358, -0.293]$); the native-minus-weights-only remainder is smaller in the median language ($+0.0138$ $[0.0093, 0.0233]$) but heterogeneous. At the individual effect-vector level the weight-term norm is larger. These engineered contrasts show that attribution depends on edit definition and endpoint, rather than identifying one universal source of damage.

Transfer and generic sensitivity. At INT6 legacy effect cosine is 0.984 and normalized error 0.165, about 1.88 INT16 reference floors. At INT4 cosine is 0.768 $[0.760, 0.776]$, error 0.664; matched RTN and GPTQ improve cosine to 0.822/0.877 and reduce error to 0.572/0.476. Drift shared with other directions indicates a substantial generic component. A post hoc common-support comparison retains 144 pairs on FLORES and 107 on Wikipedia with comparable FP effect size; the real direction's INT4 error norm is 1.14 $[1.07, 1.25]$ and 1.18 $[1.05, 1.24]$ times its swapped control. This supports a modest additional component on a selected subset, not universal feature specificity.

Averaging fingerprints increases apparent agreement relative to context-level comparison (legacy INT4 cosine gap $+0.126$). Generic sensitivity improves main-panel held-out error prediction by 21.5% $[12.7, 29.4]$ at INT4, but the INT6 gain 10.9% $[-1.2, 21.4]$ is unresolved, so the original combined-width utility criterion is not met. A similar gain at INT16 shows that the predictor is not specific to quantization. Survival and scalar-loss gradient attribution add no measurable improvement for the main eligible INT4 vector-error target. The scalar gradient nevertheless predicts its own loss target moderately ($r = 0.63$ at FP, 0.58 for the INT4 shift); an effect-vector gradient approximation was not computed.

Fresh-text check. The six-language Wikipedia panel repeats the INT4 quantizer ordering: cosine 0.754/0.808/0.876 for legacy/matched/GPTQ. Generic sensitivity improves INT4 prediction by 19/29/30%. At INT6 the criterion is met for both RTN arms but not GPTQ (3% $[-14, +17]$), whose error is near the reference. Wikipedia also supplies GPTQ calibration text from different articles, so this comparison combines method and domain effects. The panel supports portability findings on fresh text without making every main-panel label identical.

9 Discussion and Conclusion

The final RQs have constructive answers. Quantizer and activation scale shape measured preservation (RQ1). Selected groups show controlled causal relevance to multilingual next-token prediction, with convergence across ablation, graded suppression and two Gemma sizes, while comprehension and repair depend on the endpoint (RQ2). Activation and effect measurements share modest positive information but retain distinct meanings (RQ3). Intervention portability can be measured, decomposed in engineered states and improved by a better quantization recipe (RQ4).

These conclusions are meaningful within their experimental scope. A single frozen dictionary cannot establish a complete language circuit, but it can support reproducible contrasts under fixed interventions. Different dictionaries or model families test generalization; their absence from a particular arm does not erase a controlled effect. Our Llama comparison and quantizer ablations already demonstrate why that distinction matters. The work supplies an empirical measurement account rather than selective-destruction or universal circuit claims. The completed evidence stands as the present study; pending Gemma-2-9B GPTQ contributes no result.

Limitations, Challenges and Future Work

A dictionary can distribute or mix functions, and zero ablation can leave the natural activation distribution. Alternative dictionaries, layers, intervention definitions and activation quantization would test broader generalization. Activity matching does not equate every relevant feature property, and fixed groups/control draws limit the uncertainty captured by resampling. Language tiers confound script, tokenization, training exposure and competence. A reading benchmark is one endpoint among many.

Exploratory reuse of sentences limits confirmatory interpretation. Holding out languages or features does not ensure held-out documents. Numerical execution and reference perturbations matter for small differences; a paired arithmetic-precision audit remains an extension. Calibration and error compensation are not independently manipulated by the GPTQ comparison. Simulated rounded weights do not measure deployment kernels or realized compression costs. Additional independent corpora and task interventions would strengthen external validity.

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A Language Panel and Reproduction

Tier	FLORES-200 language codes
High	eng_Latn, deu_Latn, spa_Latn, fra_Latn, por_Latn, rus_Cyrl, zho_Hans, jpn_Jpan, kor_Hang
Mid	hin_Deva, ben_Beng, tam_Taml, tur_Latn, vie_Latn, ind_Latn, tha_Thai, pol_Latn, ukr_Cyrl
Low	kan_Knda, tel_Telu, yor_Latn, sun_Latn, amh_Ethi, ibo_Latn, hau_Latn, khm_Khmr, lao_Laoo

Table 2: Fixed descriptive tiers, nine languages each.

The repository is https://github.com/TH4KUR/always-late; committed evidence maps identify result files, freezes and hashes. The artifact dataset is https://huggingface.co/datasets/karma-skz/always-late-artifacts; access is private and not all item-level files are public. Run summaries are at https://wandb.ai/karma-skz-/anlp-project. The revision-4 evidence map covers the whole completed study. Earlier report wording is archived verbatim.

The frozen environment records Python 3.10, PyTorch 2.14.1, TransformerLens 3.9.0, SAE Lens 6.54.5 and Transformers 5.19.0. Most 2B jobs used 11 GB RTX 2080 Ti accelerators; the supplied 9B run used a rented 96 GB RTX PRO 6000. Checks include sham identities, restored FP weights, quantized-weight hashes and analysis reproductions. The 9B review checks raw-file provenance and headers and reproduces tables byte-identically; it does not independently repeat the supplied GPU inference. All numbers here derive from completed saved outputs; this revision introduces no model run or new statistical analysis.

B Frozen Rules and Completed Outcomes

Table 3 condenses the hypothesis inventory. Historical rules and verdicts remain in their original files. The four final RQs organize these results retrospectively; practical thresholds do not convert small positive estimates into zero effects. The course page limit motivates compression of the inventory, not omission of unsupported hypotheses.

Block	Rule or question	Completed outcome and status
Initial contrasts	Three selective-survival/tier contrasts; Holm within family	No rejection under original BOS-masked RTN. Matched/GPTQ reruns are exploratory additional families.
Association	FP establishment and peak retention	6,663/8,086 established; 70.8% retain peak at legacy INT4. Margin prediction is descriptive.
Functional F1–F4	Excess group loss, donor recovery, retained/lost subgroup effects	27/27 positive FP group effects; INT4 K64 recovery 2–3%; small subgroup gaps. Exploratory.
Belebele	Positive FP effect required before five precision statements	S0 not met; S1–S5 undefined. Original held-out family confirmatory; whole-group/prompt/positive-control checks exploratory.
Study G	H-G1/2 suppression dose/specificity; H-G3 boost dose; H-G6 consistency	Met under legacy RTN. Strict precision ordering H-G4 not met; a two-error tolerance sensitivity does not alter the original verdict.
Repair and gain	H-G5 generic/group-specific repair and inverse-shrinkage optimum	Main criteria not met. Fresh-sentence extension C meets its $1.1\times$ rule only; D shrinkage relation not supported.
State-defined G	Groups move; new groups more consequential; dose and repair	Group movement, dose and group-specific repair rules met; greater consequence and generic repair rules not met. Exploratory.
F5 and prediction	Identity, effect drift, association discrimination; continuous R_1	High identity retrieval despite drift. Binary association inconclusive/non-discriminative; continuous gain 3.1–4.4%, positive intervals, below 10% target.
Measurement audit	Gain/noise identity, scale separation, adjusted tiers and extra predictors	Identity verified; scale separation met by convention. Tier estimates mixed; gain/ν increments include zero. Exploratory.
Transfer	Fixed edit; H1 residual drift, H3 prediction, H4 averaging; group A3	Main H1/H4 supported; H3/A3 mixed; combined utility not met. External RTN utility met; GPTQ INT6 unresolved. Exploratory.
Llama	Frozen existence/dose/INT4 consistency rule	Not consistent (P1 21/27, 24 required); FP median effect positive. INT4 F5 association/prediction modestly positive. Exploratory.
Gemma-2-9B	FP effects, precision ladder, graded dependence	Positive 27/27, INT6 near FP, INT4 increase. Reversed frozen inequality and tautological selectivity item documented; no GPTQ result.

Table 3: Completed study inventory. Status belongs to each design, not to whether the estimate is positive. The numerical results and their uncertainty are in the main text and evidence map.